4.7 Change of Variables in Multiple Integrals; Jacobians

Motivation: Consider the integral: $\int_0^2 2x\sqrt{x^2+1}\,dx$. We know from calculus one since $f(x)=2x\sqrt{x^2+1}$ is continuous on $[0,2]$ then by using a u-substitution we know that if $u=x^2+1$, then $du=2x\,dx$ and the bounds become $u(0)=(0)^2+1=1$ and $u(2)=(2)^2+1=5$. Then, $\int_0^2 2x\sqrt{x^2+1}\,dx=\int_1^5 \sqrt{u}\,du=\frac{2}{3}u^{3/2}\Big|_1^5=\frac{2}{3}\left(5^{3/2}-1\right)$.

In this process we see that a few things have changed in order to complete the substitution:

(1) The bounds (i.e., the interval we are integrating over).
(2) The function we are integrating.
(3) The differential dx had to be changed by finding the relationship between dx & du.

In Calculus III we have been integrating functions of two variables and we wish to see how we might be able to perform a *uv*-substitution (what we more commonly call a *change of variables*). When doing this our substitution will require that we change similar features of our integral.

(1) The region that we are integrating over.
(2) The function we are integrating.
(3) The differential dA_{xy} (*dxdy* or *dydx*) will need to be changed to dA_{uv} (*dudv* or *dvdu*) by finding the relationship between dA_{xy} & dA_{uv}

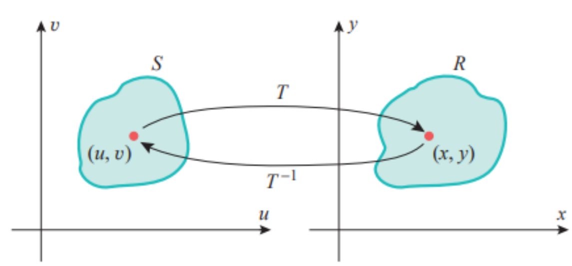
Before we look at how each of these changes work within a given *uv*-substitution we first introduce some important terminology for this section.

Definition: A transformation from the *uv*-plane to the *xy*-plane is a function of the form $T(u,v)=\left(g(u,v),h(u,v)\right)$. Intuitively, we think of $g(u,v)$ as giving us an *x* value, and we think of $h(u,v)$ as giving us a *y* value.

Note: In function mapping notation we will write $T:\mathbb{R}^2\to\mathbb{R}^2$ since T takes in an ordered pair and spits out an ordered pair.

Definition: If $T(a,b)=(c,d)$ we say that (*c*, *d*) is the image of the point (*a*, *b*) under the transformation *T*. If no two points have the same image, *T* is called one-to-one.

Note: If T is one-to-one this implies that T has an inverse transformation we call T^{-1} which maps points from the xy-plane to the uv-plane.



Example 1: Consider the transformation given by $T(u,v)=(u+v,v^3)$.

a) Express the substitutions this transformation implies.

b) What is the image of (2,3) under T?

c) Suppose that $(0,0)$ is a point in the uv-plane, where does T map this point in xy-plane?

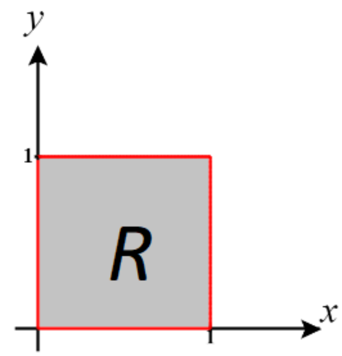
d) Is T a one-to-one transformation?

Now we turn our attention to how uv-substitutions affect the region we are integrating over.

Example 2: Consider the transformation given by $T(u,v)=(u^2-v^2,2uv)$. Find the image of the following square in the uv-plane: $S=\{(u,v)|0\le u\le 1, 0\le v\le 1\}$ under the transformation T.

Fact: If in the transformation $T(u,v)=(f(u,v),g(u,v))$ both $f(u,v)$ and $g(u,v)$ are linear functions in u and v then the transformation T maps lines to lines OR lines to points. The same holds for T^{-1}.

Example 3: Suppose we are going to work with a substitution where $u=3x-2y$ & $v=x+y$ (note this is a T^{-1} transformation). If this substitution is used on a region R in the xy-plane, shown below, it will create a region S in the uv-plane. Determine the region S in the uv-plane.



The past several examples have shown us how given transformations affect the region of integration.

Now the question is: How does our new region relate to the differentials that we start with and want to switch to and what does this imply about the areas of the regions that we have?

Example 4: Suppose that we are integrating $\iint_R 1\,dA$ where R is a closed region with boundary given by the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$.

a) Show how the substitution given by the transformation $T(u,v)=(3u,2v)$ gives us an easier region to integrate over.

b) Determine the relationship between dA_{xy} & dA_{uv} base on this substitution.

c) Compute $\iint_R 1\,dA$ by using the given *uv*-substitution.

Note: It is not always the case that the area of R will scale uniformly. The next example shows this.

Example 5: Suppose we are going to work with a substitution where $u = \frac{x^3}{3}$ & $v = \frac{y^2}{2}$ over a region *R*. Determine the relationship between the area *R* and its transformation *S*.

Note: It is also possible that finding such a relationship between $dudv$ & $dxdy$ *is not so simple as in the previous two examples. For example, consider if we worked with a substitution like* $u = 3x - 2y$ & $v = x + y$.

This leads us to ask: Is there a generalized way of determining this "scaling factor"?

Definition: The Jacobian of the transformation *T* given by $T(u,v) = (x(u,v), y(u,v))$ is

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial y}{\partial u}\frac{\partial x}{\partial v}$$ (assuming these partial derivatives exist).

Note: An exploration helping to explain this idea is included in the full lecture notes in Blackboard.

Note: It can be useful to think of the Jacobian as consisting of the gradient of the x function in its first row and the gradient of the y function in its second row.

Note: Intuitively we can think of the Jacobian as an "area correction/scaling factor". As it turns out the Jacobian appears when we make a change of variables in a double integral.

Example 6: Find the Jacobian of the transformation given by: $T(r,\theta) = (r\cos(\theta), r\sin(\theta))$.

Theorem (Change of Variables in a Double Integral):
Suppose that T is a transformation with continuous first-order partials, whose Jacobian is nonzero and that T maps a region S in the uv-plane onto a region R in the xy-plane. Suppose that f is continuous on R and that R and S are type I or type II plane regions. Suppose that T is one-to-one, except perhaps on the boundary of S. Then,

$$\iint_R f(x,y)\,dA = \iint_S f(x(u,v), y(u,v)) \left|\frac{\partial(x,y)}{\partial(u,v)}\right| dudv .$$

Note: The precise statement of conditions under which this formula works is beyond the scope of this course. The formula However, the theorem intuitively makes sense based upon our exploration above.

Note: We can also apply the above theorem when R and S can be decomposed into type 1 and type 2 regions (though this may require the use of more than one double integral). Also note that the order of the differentials on the right side of the equation may be taken to be dvdu.

Example 7: Suppose that $R = \{(x,y) | 0 \le x \le 1 \ \&\ 0 \le y \le 1\}$. Compute $\iint_R x^2 y\, dA$ by changing $u = x \ \&\ v = xy$.

Example 8: In the last example we had to solve for x and y in order to find the Jacobian to complete the change of variables. Compute $\dfrac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} \\ \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial y} \end{vmatrix}$. How does this compare to the Jacobian?

Remark: The product $\frac{\partial(x,y)}{\partial(u,v)} \cdot \frac{\partial(u,v)}{\partial(x,y)} = 1$. This mean that the Jacobian and $\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$ are inverses of one another (assuming these partial derivatives exist).

Proof (Shown in the lecture notes).

Remark: If $dxdy = |J|\,dudv$ then $\frac{1}{|J|}dxdy = dudv$

Proof (follows from the above remark).

Note that these recent facts imply that to determine the scale factor between the regions R and S we can always just compute J or $\frac{1}{J}$, whichever is easiest!

We can now give a rough outline of the steps we take when performing a change of variables in a double integral.

Steps for Performing a Change of Variables in a Double Integral:

1. Let $x = x(u,v)$ and $y = y(u,v)$ OR $u = u(x,y)$ and $v = v(x,y)$ (sometimes we will have to determine these formulas, and sometimes these formulas will be given).
2. Draw a picture of the region *R* of integration and give inequalities that determine the region *R*
3. Ensure that the Transformation *T (or* T^{-1}*)* determined by the equations in 1 is a transformation with continuous first order partials and calculate the Jacobian (or the inverse Jacobian).
4. Determine the region *S* in the *uv*-plane. It may be helpful to solve the equations in 1 for *x* and *y* first.
5. Verify the hypotheses of the change of variable theorem are met (i.e. the Jacobian of *T* is nonzero, the function of integration is continuous on *R*, and *T* is one-to-one except perhaps on the boundary of *S*).
6. Apply the change of variables formula: $\iint_R f(x,y)\,dA = \iint_S f(x(u,v), y(u,v)) \left|\frac{\partial(x,y)}{\partial(u,v)}\right| dudv$.

Note: In MTH 202 we will not always write out all the details for step 5, but we will at least mentally verify the hypotheses.

Example 9: Consider the double integral $\iint_R y\,dA$ where $R = \left\{ (x,y) \middle| \frac{y^2}{4} - 1 \le x \le 1 - \frac{y^2}{4}, y \ge 0 \right\}$. Use the change of variables $x = u^2 - v^2 \quad y = 2uv$ to find this double integral.

Note: In the past problems we have been given a uv-substitution to use, but in general this is not always the case. Much like in Calculus I we must find our own substitutions to use. Our main goals when doing this are to:

1. *Make sure our integrand is easy to work with (we have seen examples of how integrands setup one way may be more difficult than in others).*
2. *Make sure that the region that we are integrating over is easy to work with (we have seen examples of moving to easier regions in polar, spherical, and cylindrical coordinates).*

Example 10: Consider the integral $\iint_R e^{(x+y)/(x-y)}dA$ where R is the trapezoidal region with vertices (1,0), (2,0), (0,-2), and (0,-1).

a) What is a possible change of variables that will make this integral easier to evaluate (give formulas for x and y)?

b) Use the change of variables from (a) to evaluate the integral.

We end 4.7 by briefly discussing how the change of variables theorem generalizes to triple integrals.

Definition: Suppose that T is a transformation from uvw-space to xyz-space given by $T(u,v,w)=(g(u,v,w),h(u,v,w),k(u,v,w))$. If the first order partials of the component functions of T are continuous, then the <u>Jacobian</u> of T is the three-by-three determinant:

$$\frac{\partial(x,y,z)}{\partial(u,v,w)}=\begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}.$$

Fact: Under similar hypotheses to the change of variables for a double integral theorem, it is possible to prove a theorem for the change of variables of a triple integral. The formula for the change of variables is:

$$\iiint_B f(x,y,z)\,dV=\iiint_S f(x(u,v,w),y(u,v,w),z(u,v,w))\left|\frac{\partial(x,y,z)}{\partial(u,v,w)}\right|dudvdw.$$

<u>**Example 11:**</u> Suppose that E is the region in 3-space given in spherical by $E=\{(\rho,\theta,\phi):\alpha\le\theta\le\beta,\ c\le\phi\le d,\ g_1(\theta,\phi)\le\rho\le g_2(\theta,\phi)\}$.

Then, E is the image of the region $S=\{(\rho,\theta,\phi):\alpha\le\theta\le\beta,\ c\le\phi\le d,\ g_1(\theta,\phi)\le\rho\le g_2(\theta,\phi)\}$ in the $\rho\theta\phi-$space under the transformation $T(\rho,\theta,\phi)=(\rho\sin(\phi)\cos(\theta),\rho\sin(\phi)\sin(\theta),\rho\cos(\phi))$.

Use the change of variables fact just mentioned to find a formula for $\iiint_E f(x,y,z)\,dV$ (you may assume f is continuous on E).